EST II — MATH LEVEL 2 • UNIT 3.01

C Polynomial, Rational, Exponential & Logarithmic Functions

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

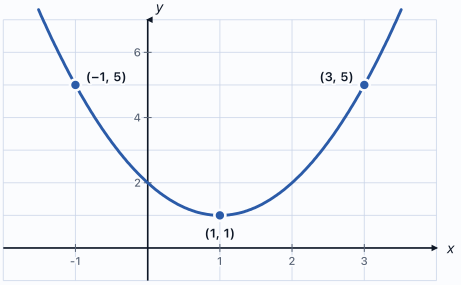
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 D	Q2 A	Q3 D	Q4 D	Q5 D
Q6 C	Q7 A	Q8 B	Q9 C	Q10 B
Q11 A	Q12 D	Q13 B	Q14 A	Q15 B
Q16 A	Q17 B	Q18 D	Q19 C	Q20 B
Q21 D	Q22 D	Q23 C	Q24 D	Q25 A

Polynomial, Rational, Exponential & Logarithmic Functions Practice Exercises

1 MCQ

The graph of the function $y = f(x) = ax^2 + bx + c$ is shown below. The graph passes through the points $(-1, 5)$, $(1, 1)$, and $(3, 5)$. Determine the value of b.



A 2

B 1

C 0

D -2 ✓

WORKED SOLUTION & VERIFICATION

The general form of the quadratic function is $y = ax^2 + bx + c$. We are given three points that lie on the graph: $(-1, 5)$, $(1, 1)$, and $(3, 5)$.

Substituting these points into the equation, we get a system of three linear equations with three unknowns (a, b, and c):

1. For $(-1, 5)$: $a(-1)^2 + b(-1) + c = 5 \Rightarrow a - b + c = 5$
2. For $(1, 1)$: $a(1)^2 + b(1) + c = 1 \Rightarrow a + b + c = 1$
3. For $(3, 5)$: $a(3)^2 + b(3) + c = 5 \Rightarrow 9a + 3b + c = 5$

We can solve this system of equations. Let's subtract equation (1) from equation (2):

$(a + b + c) - (a - b + c) = 1 - 5$

$2b = -4$

$b = -2$

Alternatively, notice that the y-values for $x = -1$ and $x = 3$ are the same (both are 5). This indicates that the axis of symmetry of the parabola is exactly in the middle of $x = -1$ and $x = 3$. The x-coordinate of the vertex (axis of symmetry) is given by $-\frac{b}{2a}$.

Axis of symmetry: $x = \frac{-1+3}{2} = \frac{2}{2} = 1$

So, $-\frac{b}{2a} = 1$, which implies $b = -2a$.

Now, substitute this into equation (2): $a + b + c = 1$

$a + (-2a) + c = 1$

$-a + c = 1 \Rightarrow c = a + 1$

Substitute $b = -2a$ and $c = a + 1$ into equation (1):

$a - (-2a) + (a + 1) = 5$

$a + 2a + a + 1 = 5$

$4a + 1 = 5$

$4a = 4$

$a = 1$

Now we can find b using $b = -2a$:

$b = -2(1)$

$b = -2$

To confirm, we can also find c: $c = a + 1 = 1 + 1 = 2$. The function is $y = x^2 - 2x + 2$.

Let's check the points:

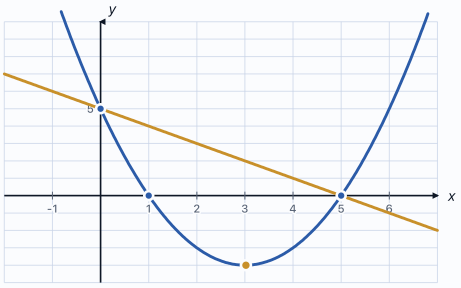
For $(-1, 5)$: $(-1)^2 - 2(-1) + 2 = 1 + 2 + 2 = 5$ (Correct)

For $(1, 1)$: $(1)^2 - 2(1) + 2 = 1 - 2 + 2 = 1$ (Correct)

For $(3, 5)$: $(3)^2 - 2(3) + 2 = 9 - 6 + 2 = 5$ (Correct)
The value of b is -2.

2 MCQ

Consider the functions $f(x) = x^2 - 6x + 5$ and $g(x) = -x + 5$. The graphs of these two functions intersect at two points. Let these points be A and B. A third function, $h(x)$, is defined as the difference between the x-coordinates of these intersection points, i.e., $h(x) = |x_A - x_B|$. What is the value of $h(x)$?



A $\sqrt{25}$ ✓

B $\sqrt{16}$

C $\sqrt{36}$

D $\sqrt{49}$

WORKED SOLUTION & VERIFICATION

To find the intersection points of the graphs of $f(x)$ and $g(x)$, we set $f(x) = g(x)$.
$x^2 - 6x + 5 = -x + 5$
Subtract $-x + 5$ from both sides:
$x^2 - 6x + 5 - (-x + 5) = 0$
$x^2 - 6x + 5 + x - 5 = 0$
Combine like terms:
$x^2 - 5x = 0$
Factor out x:
$x(x - 5) = 0$
This gives two possible x-values for the intersection points:
$x_1 = 0$
$x_2 = 5$
Let $x_A = 0$ and $x_B = 5$ (or vice versa). The problem defines $h(x)$ as the absolute difference between these x-coordinates: $h(x) = |x_A - x_B|$.
$h(x) = |0 - 5| = |-5| = 5$.
We are asked for the value of $h(x)$, which is 5. We need to express this value in a form that matches the choices. The choices are given as square roots.
$\sqrt{16} = 4$
$\sqrt{25} = 5$
$\sqrt{36} = 6$
$\sqrt{49} = 7$
The value of $h(x)$ is 5, which corresponds to $\sqrt{25}$.

3 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. What are the coordinates of the vertex of the parabola represented by this function?

A (2, 1)

B (1, 0)

C (3, 0)

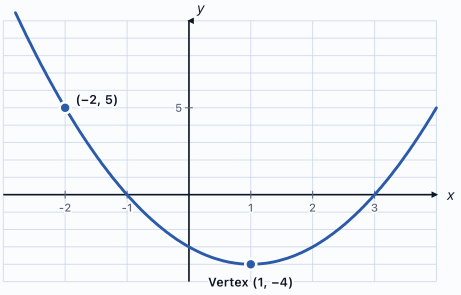
D **(2, -1)** ✓

WORKED SOLUTION & VERIFICATION

The given function is a quadratic function in the form $f(x) = ax^2 + bx + c$, where $a = 1$, $b = -4$, and $c = 3$. The x-coordinate of the vertex of a parabola is given by the formula $x = -\frac{b}{2a}$. Substituting the values, we get $x = -\frac{-4}{2(1)} = \frac{4}{2} = 2$. To find the y-coordinate of the vertex, we substitute this x-value back into the function: $f(2) = (2)^2 - 4(2) + 3 = 4 - 8 + 3 = -1$. Therefore, the coordinates of the vertex are (2, -1).

4 MCQ

The graph of the quadratic function $f(x) = x^2 + bx + c$ is shown below. The graph passes through the point $(-2, 5)$ and has its vertex at $(1, -4)$. Determine the value of b.



A 1

B -1

C 2

D **-2** ✓

WORKED SOLUTION & VERIFICATION

The vertex of a parabola given by $f(x) = ax^2 + bx + c$ is at $x = -\frac{b}{2a}$. In this case, $a = 1$ and the x-coordinate of the vertex is given as 1. So, $1 = -\frac{b}{2(1)}$. This simplifies to $1 = -\frac{b}{2}$, which means $b = -2$.

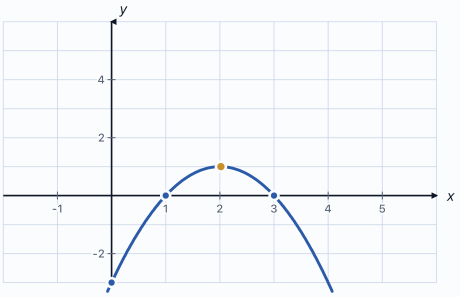
Alternatively, we can use the vertex form of a quadratic equation: $f(x) = a(x - h)^2 + k$, where (h, k) is the vertex. Here, $a = 1$ and $(h, k) = (1, -4)$, so $f(x) = (x - 1)^2 - 4$. Expanding this, we get $f(x) = (x^2 - 2x + 1) - 4 = x^2 - 2x - 3$. Comparing this to $f(x) = x^2 + bx + c$, we see that $b = -2$ and $c = -3$.

We can verify that the point $(-2, 5)$ lies on this parabola: $f(-2) = (-2)^2 - 2(-2) - 3 = 4 + 4 - 3 = 5$. This confirms our values.

Common errors might involve misinterpreting the vertex formula or making sign errors during expansion.

5 MCQ

The graph of the function $f(x) = -x^2 + 4x - 3$ is shown below. Determine the y-coordinate of the vertex of the parabola.



A 3

B 4

C 2

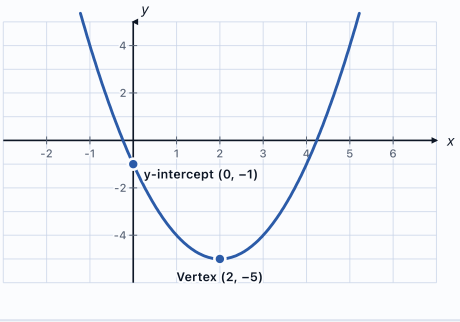
D 1 ✓

WORKED SOLUTION & VERIFICATION

The vertex of a parabola in the form $f(x) = ax^2 + bx + c$ occurs at $x = -\frac{b}{2a}$. In this case, $a = -1$ and $b = 4$. So, the x-coordinate of the vertex is $x = -\frac{4}{2(-1)} = -\frac{4}{-2} = 2$. To find the y-coordinate of the vertex, substitute this x-value back into the function: $f(2) = -(2)^2 + 4(2) - 3 = -4 + 8 - 3 = 1$. Therefore, the vertex is at (2, 1) and the y-coordinate is 1.

6 MCQ

The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown below. The vertex of the parabola is at $(2, -5)$ and the y-intercept is at $(0, -1)$. Determine the value of a.



A -1

B 2

C 1 ✓

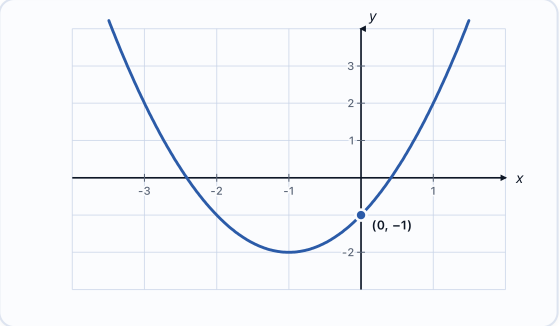
D -2

WORKED SOLUTION & VERIFICATION

The general form of a quadratic function is $f(x) = ax^2 + bx + c$. The vertex form of a quadratic function is $f(x) = a(x - h)^2 + k$, where (h, k) is the vertex. Given the vertex is $(2, -5)$, the function can be written as $f(x) = a(x - 2)^2 - 5$. We are also given that the y-intercept is $(0, -1)$. This means that when $x = 0, f(x) = -1$. Substituting these values into the vertex form: $-1 = a(0 - 2)^2 - 5$. This simplifies to $-1 = a(-2)^2 - 5$, which is $-1 = 4a - 5$. Adding 5 to both sides gives $4 = 4a$. Dividing by 4 yields $a = 1$.

7 MCQ

Consider the graph of the function $f(x)$ shown below. What is the value of $f(0)$?



- **A** -1 ✓
- **B** 1
- **C** 0
- **D** 2

WORKED SOLUTION & VERIFICATION

The value of $f(0)$ is the y-coordinate of the point where the graph of the function intersects the y-axis. From the graph, we can see that the function intersects the y-axis at the point $(0, -1)$. Therefore, $f(0) = -1$.

8 MCQ

If $f(x) = x^2$ and $g(x) = 2f(x - 1) + 8$, find $g(6)$.

- **A** 80
- **B** 58 ✓
- **C** 56
- **D** 60

WORKED SOLUTION & VERIFICATION

Applying Functions & Graphs principles, $g(6) = 2(6 - 1)^2 + 8 = 58$. Therefore, the verified correct option is 58.

9 MCQ

If $f(x) = x^2$ and $g(x) = 1f(x - 2) - 6$, find $g(-4)$.

- **A** 31
- **B** 10
- **C** 30 ✓
- **D** 28

WORKED SOLUTION & VERIFICATION

Applying Functions & Graphs principles, $g(-4) = 1(-4 - 2)^2 - 6 = 30$. Therefore, the verified correct option is 30.

10 MCQ

If $f(x) = x^2$ and $g(x) = 3f(x - 6) - 5$, find $g(4)$.

- **A** 43
- **B** 7 ✓
- **C** 5
- **D** 10

WORKED SOLUTION & VERIFICATION

Applying Functions & Graphs principles, $g(4) = 3(4 - 6)^2 - 5 = 7$. Therefore, the verified correct option is 7.

11 MCQ

Suppose that $f(x) = x^2$ and $g(x) = 2f(x-4) + 1$, find $g(3)$.

- **A** 3 ✓
- **B** 1
- **C** 5
- **D** 19

WORKED SOLUTION & VERIFICATION

Applying Functions & Graphs principles, $g(3) = 2(3-4)^2 + 1 = 3$. Therefore, the verified correct option is 3.

12 MCQ

Assuming that $f(x) = x^2$ and $g(x) = 3f(x-3) + 6$, find $g(3)$.

- **A** 9
- **B** 33
- **C** 4
- **D** 6 ✓

WORKED SOLUTION & VERIFICATION

Applying Functions & Graphs principles, $g(3) = 3(3-3)^2 + 6 = 6$. Therefore, the verified correct option is 6.

13 MCQ

Under the condition that $f(x) = x^2$ and $g(x) = 1f(x-6) + 7$, find $g(9)$.

- **A** 17
- **B** 16 ✓
- **C** 88
- **D** 14

WORKED SOLUTION & VERIFICATION

Applying Functions & Graphs principles, $g(9) = 1(9-6)^2 + 7 = 16$. Therefore, the verified correct option is 16.

14 MCQ

Suppose that $f(x) = x^2$ and $g(x) = 1f(x-6) - 3$, find $g(2)$.

- **A** 13 ✓
- **B** 11
- **C** 14
- **D** 1

WORKED SOLUTION & VERIFICATION

Applying Functions & Graphs principles, $g(2) = 1(2-6)^2 - 3 = 13$. Therefore, the verified correct option is 13.

15 MCQ

If $f(x) = x^2$ and $g(x) = -1f(x-6) - 5$, find $g(4)$.

- **A** -10
- **B** -9 ✓
- **C** -21
- **D** -11

WORKED SOLUTION & VERIFICATION

Applying Functions & Graphs principles, $g(4) = -1(4-6)^2 - 5 = -9$. Therefore, the verified correct option is -9.

16 MCQ

Provided that $f(x) = x^2$ and $g(x) = 1f(x-1) + 4$, find $g(9)$.

A 68 ✓	B 69
C 85	D 66

WORKED SOLUTION & VERIFICATION

Applying Functions & Graphs principles, $g(9) = 1(9-1)^2 + 4 = 68$. Therefore, the verified correct option is 68.

17 MCQ

Assuming that $f(x) = x^2$ and $g(x) = 3f(x-4) - 2$, find $g(1)$.

A 28	**B** 25 ✓
C 1	D 23

WORKED SOLUTION & VERIFICATION

Applying Functions & Graphs principles, $g(1) = 3(1-4)^2 - 2 = 25$. Therefore, the verified correct option is 25.

18 MCQ

Under the condition that $f(x) = x^2$ and $g(x) = 1f(x-5) - 3$, find $g(5)$.

A -5	B -2
C 22	**D** -3 ✓

WORKED SOLUTION & VERIFICATION

Applying Functions & Graphs principles, $g(5) = 1(5-5)^2 - 3 = -3$. Therefore, the verified correct option is -3.

19 MCQ

If $f(x) = x^2$ and $g(x) = -1f(x-5) - 6$, find $g(-3)$.

A -15	B -72
C -70 ✓	D -71

WORKED SOLUTION & VERIFICATION

Applying Functions & Graphs principles, $g(-3) = -1(-3-5)^2 - 6 = -70$. Therefore, the verified correct option is -70.

20 MCQ

Suppose that $f(x) = x^2$ and $g(x) = 3f(x-6) - 7$, find $g(-1)$.

A 138	B **140** ✓
C 143	D -4

WORKED SOLUTION & VERIFICATION

Applying Functions & Graphs principles, $g(-1) = 3(-1-6)^2 - 7 = 140$. Therefore, the verified correct option is 140.

21 MCQ

Find the domain of $f(x) = \sqrt{x+9}$.

A $x \leq -9$	B x>-9
C $x \neq -9$	D $x \geq -9$ ✓

WORKED SOLUTION & VERIFICATION

Applying Functions & Graphs principles, Need $x + 9 \geq 0$, so $x \geq -9$. Therefore, the verified correct option is $x \geq -9$.

22 MCQ

Find the domain of $f(x) = \sqrt{x-8}$.

A $x \leq 8$	B x>8
C $x \neq 8$	D $x \geq 8$ ✓

WORKED SOLUTION & VERIFICATION

Applying Functions & Graphs principles, Need $x - 8 \geq 0$, so $x \geq 8$. Therefore, the verified correct option is $x \geq 8$.

23 MCQ

Calculate the domain of $f(x) = \sqrt{x-4}$.

A $x \leq 4$	B x>4
C $x \geq 4$ ✓	D $x \neq 4$

WORKED SOLUTION & VERIFICATION

Applying Functions & Graphs principles, Need $x - 4 \geq 0$, so $x \geq 4$. Therefore, the verified correct option is $x \geq 4$.

24 MCQ

Compute the domain of $f(x) = \sqrt{x - 0}$.

- A $x \leq 0$
- B x>0
- C $x \neq 0$
- D $x \geq 0$ ✓

WORKED SOLUTION & VERIFICATION

Applying Functions & Graphs principles, Need $x - 0 \geq 0$, so $x \geq 0$. Therefore, the verified correct option is $x \geq 0$.

25 MCQ

Calculate the domain of $f(x) = \sqrt{x + 1}$.

- A $x \geq -1$ ✓
- B $x \leq -1$
- C x>-1
- D $x \neq -1$

WORKED SOLUTION & VERIFICATION

Applying Functions & Graphs principles, Need $x + 1 \geq 0$, so $x \geq -1$. Therefore, the verified correct option is $x \geq -1$.